$C^{(1)\,i1\_i2\_i3\_i4\_}_{lq} \to$

$$\frac{1}{16\pi^2}\Big(-\frac{1}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde d}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{216}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde d}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde d}^{p}]$$

$$\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{27}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde e}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{72}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde e}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{4}{135}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde e}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{54}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde l}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{5}{144}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde l}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{2}{135}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde l}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{162}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde q}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{5}{432}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde q}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{2}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde q}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{81}\sum_p g_1^4\,\mathrm{LF}_{3,0}[m_{\tilde u}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{5}{54}\sum_p g_1^4\,\mathrm{LF}_{4,-1}[m_{\tilde u}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{16}{405}\sum_p g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde u}^{p}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{54}g_1^4\,\mathrm{LF}_{3,0}[\tilde\mu]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{36}g_1^4\,\mathrm{LF}_{4,-1}[\tilde\mu]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{4}{135}g_1^4\,\mathrm{LF}_{5,-2}[\tilde\mu]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{288}g_1^4\,\mathrm{LF}_{2,1,0}[m_1,m_{\tilde l}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{288}g_1^4\,\mathrm{LF}_{2,2,-1}[m_1,m_{\tilde l}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{144}g_1^4\,\mathrm{LF}_{3,1,-1}[m_1,m_{\tilde l}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{288}g_1^4\,\mathrm{LF}_{4,1,-2}[m_1,m_{\tilde l}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{288}g_1^4\,\mathrm{LF}_{2,1,0}[m_1,m_{\tilde l}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{288}g_1^4\,\mathrm{LF}_{2,2,-1}[m_1,m_{\tilde l}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{144}g_1^4\,\mathrm{LF}_{3,1,-1}[m_1,m_{\tilde l}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{288}g_1^4\,\mathrm{LF}_{4,1,-2}[m_1,m_{\tilde l}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{2592}g_1^4\,\mathrm{LF}_{2,1,0}[m_1,m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{2592}g_1^4\,\mathrm{LF}_{2,2,-1}[m_1,m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{1296}g_1^4\,\mathrm{LF}_{3,1,-1}[m_1,m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{2592}g_1^4\,\mathrm{LF}_{4,1,-2}[m_1,m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{2592}g_1^4\,\mathrm{LF}_{2,1,0}[m_1,m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{2592}g_1^4\,\mathrm{LF}_{2,2,-1}[m_1,m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{1296}g_1^4\,\mathrm{LF}_{3,1,-1}[m_1,m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{2592}g_1^4\,\mathrm{LF}_{4,1,-2}[m_1,m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{2,1,0}[m_2,m_{\tilde l}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{2,2,-1}[m_2,m_{\tilde l}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{48}g_1^2g_2^2\,\mathrm{LF}_{3,1,-1}[m_2,m_{\tilde l}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{4,1,-2}[m_2,m_{\tilde l}^{i1}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{2,1,0}[m_2,m_{\tilde l}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{2,2,-1}[m_2,m_{\tilde l}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{48}g_1^2g_2^2\,\mathrm{LF}_{3,1,-1}[m_2,m_{\tilde l}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{4,1,-2}[m_2,m_{\tilde l}^{i2}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{2,1,0}[m_2,m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{2,2,-1}[m_2,m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{48}g_1^2g_2^2\,\mathrm{LF}_{3,1,-1}[m_2,m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{4,1,-2}[m_2,m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{2,1,0}[m_2,m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{2,2,-1}[m_2,m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{48}g_1^2g_2^2\,\mathrm{LF}_{3,1,-1}[m_2,m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{4,1,-2}[m_2,m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{54}g_1^2g_3^2\,\mathrm{LF}_{2,1,0}[m_3,m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{54}g_1^2g_3^2\,\mathrm{LF}_{2,2,-1}[m_3,m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{3,1,-1}[m_3,m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{54}g_1^2g_3^2\,\mathrm{LF}_{4,1,-2}[m_3,m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{54}g_1^2g_3^2\,\mathrm{LF}_{2,1,0}[m_3,m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{54}g_1^2g_3^2\,\mathrm{LF}_{2,2,-1}[m_3,m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{3,1,-1}[m_3,m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{54}g_1^2g_3^2\,\mathrm{LF}_{4,1,-2}[m_3,m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{18}g_1^2\,\overline{y_d}^{i4p}\,y_d^{i3p}\,\mathrm{LF}_{2,1,0}[m_{\tilde d}^{p},\tilde\mu]\,\delta_{i1i2}+\frac{1}{36}g_1^2\,\overline{y_d}^{i4p}\,y_d^{i3p}\,\mathrm{LF}_{2,2,-1}[m_{\tilde d}^{p},\tilde\mu]\,\delta_{i1i2}+$$

$$\frac{1}{36}g_1^2\,\overline{y_d}^{i4p}\,y_d^{i3p}\,\mathrm{LF}_{3,1,-1}[m_{\tilde d}^{p},\tilde\mu]\,\delta_{i1i2}+\frac{1}{18}g_1^2\,\overline{y_e}^{i2p}\,y_e^{i1p}\,\mathrm{LF}_{2,1,0}[m_{\tilde e}^{p},\tilde\mu]\,\delta_{i3i4}-$$

$$\frac{1}{36}g_1^2\,\overline{y_e}^{i2p}\,y_e^{i1p}\,\mathrm{LF}_{2,2,-1}[m_{\tilde e}^{p},\tilde\mu]\,\delta_{i3i4}-\frac{1}{36}g_1^2\,\overline{y_e}^{i2p}\,y_e^{i1p}\,\mathrm{LF}_{3,1,-1}[m_{\tilde e}^{p},\tilde\mu]\,\delta_{i3i4}+$$

$$\frac{1}{144}g_1^4\,\mathrm{LF}_{2,1,0}[m_{\tilde l}^{i1},m_1]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{288}g_1^4\,\mathrm{LF}_{3,1,-1}[m_{\tilde l}^{i1},m_1]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{48}g_1^2g_2^2\,\mathrm{LF}_{2,1,0}[m_{\tilde l}^{i1},m_2]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde l}^{i1},m_2]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{144}g_1^4\,\mathrm{LF}_{2,1,0}[m_{\tilde l}^{i2},m_1]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{288}g_1^4\,\mathrm{LF}_{3,1,-1}[m_{\tilde l}^{i2},m_1]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{48}g_1^2g_2^2\,\mathrm{LF}_{2,1,0}[m_{\tilde l}^{i2},m_2]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde l}^{i2},m_2]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{1296}g_1^4\,\mathrm{LF}_{2,1,0}[m_{\tilde q}^{i3},m_1]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{2592}g_1^4\,\mathrm{LF}_{3,1,-1}[m_{\tilde q}^{i3},m_1]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{48}g_1^2g_2^2\,\mathrm{LF}_{2,1,0}[m_{\tilde q}^{i3},m_2]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde q}^{i3},m_2]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{2,1,0}[m_{\tilde q}^{i3},m_3]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{54}g_1^2g_3^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde q}^{i3},m_3]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{1296}g_1^4\,\mathrm{LF}_{2,1,0}[m_{\tilde q}^{i4},m_1]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{2592}g_1^4\,\mathrm{LF}_{3,1,-1}[m_{\tilde q}^{i4},m_1]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{48}g_1^2g_2^2\,\mathrm{LF}_{2,1,0}[m_{\tilde q}^{i4},m_2]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}g_1^2g_2^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde q}^{i4},m_2]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{27}g_1^2g_3^2\,\mathrm{LF}_{2,1,0}[m_{\tilde q}^{i4},m_3]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{54}g_1^2g_3^2\,\mathrm{LF}_{3,1,-1}[m_{\tilde q}^{i4},m_3]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{9}g_1^2\,\overline{y_u}^{i4p}\,y_u^{i3p}\,\mathrm{LF}_{2,1,0}[m_{\tilde u}^{p},\tilde\mu]\,\delta_{i1i2}-\frac{1}{18}g_1^2\,\overline{y_u}^{i4p}\,y_u^{i3p}\,\mathrm{LF}_{2,2,-1}[m_{\tilde u}^{p},\tilde\mu]\,\delta_{i1i2}-$$

$$\frac{1}{18}g_1^2\,\overline{y_u}^{i4p}\,y_u^{i3p}\,\mathrm{LF}_{3,1,-1}[m_{\tilde u}^{p},\tilde\mu]\,\delta_{i1i2}+\frac{1}{36}g_1^2\,\overline{y_d}^{i4p}\,y_d^{i3p}\,\mathrm{LF}_{2,1,0}[\tilde\mu,m_{\tilde d}^{p}]\,\delta_{i1i2}+$$

$$\frac{5}{72}g_1^2\,\overline{y_d}^{i4p}\,y_d^{i3p}\,\mathrm{LF}_{3,1,-1}[\tilde\mu,m_{\tilde d}^{p}]\,\delta_{i1i2}-\frac{1}{72}g_1^2\,\overline{y_d}^{i4p}\,y_d^{i3p}\,\mathrm{LF}_{4,1,-2}[\tilde\mu,m_{\tilde d}^{p}]\,\delta_{i1i2}-$$

$$\frac{1}{36}g_1^2\,\overline{y_e}^{i2p}\,y_e^{i1p}\,\mathrm{LF}_{2,1,0}[\tilde\mu,m_{\tilde e}^{p}]\,\delta_{i3i4}+\frac{1}{72}g_1^2\,\overline{y_e}^{i2p}\,y_e^{i1p}\,\mathrm{LF}_{3,1,-1}[\tilde\mu,m_{\tilde e}^{p}]\,\delta_{i3i4}-$$

$$\frac{1}{72}g_1^2\,\overline{y_e}^{i2p}\,y_e^{i1p}\,\mathrm{LF}_{4,1,-2}[\tilde\mu,m_{\tilde e}^{p}]\,\delta_{i3i4}-\frac{1}{18}g_1^2\,\overline{y_u}^{i4p}\,y_u^{i3p}\,\mathrm{LF}_{2,1,0}[\tilde\mu,m_{\tilde u}^{p}]\,\delta_{i1i2}-$$

$$\frac{1}{72}g_1^2\,\overline{y_u}^{i4p}\,y_u^{i3p}\,\mathrm{LF}_{3,1,-1}[\tilde\mu,m_{\tilde u}^{p}]\,\delta_{i1i2}-\frac{1}{72}g_1^2\,\overline{y_u}^{i4p}\,y_u^{i3p}\,\mathrm{LF}_{4,1,-2}[\tilde\mu,m_{\tilde u}^{p}]\,\delta_{i1i2}+$$

$$\frac{1}{288}g_1^4\,\mathrm{LF}_{2,1,1,-1}[m_1,m_{\tilde l}^{i1},m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{144}g_1^4\,m_1^2\,\mathrm{LF}_{2,1,1,0}[m_1,m_{\tilde l}^{i1},m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{288}g_1^4\,\mathrm{LF}_{2,1,1,-1}[m_1,m_{\tilde l}^{i2},m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{144}g_1^4\,m_1^2\,\mathrm{LF}_{2,1,1,0}[m_1,m_{\tilde l}^{i2},m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{3}{32}g_2^4\,\mathrm{LF}_{2,1,1,-1}[m_2,m_{\tilde l}^{i1},m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{3}{16}g_2^4\,m_2^2\,\mathrm{LF}_{2,1,1,0}[m_2,m_{\tilde l}^{i1},m_{\tilde q}^{i3}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{3}{32}g_2^4\,\mathrm{LF}_{2,1,1,-1}[m_2,m_{\tilde l}^{i2},m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{3}{16}g_2^4\,m_2^2\,\mathrm{LF}_{2,1,1,0}[m_2,m_{\tilde l}^{i2},m_{\tilde q}^{i4}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{8}\,\overline{y_d}^{i4p}\,y_d^{i3p}\,\overline{y_e}^{i2r}\,y_e^{i1r}\,\mathrm{LF}_{2,1,1,-1}[\tilde\mu,m_{\tilde d}^{p},m_{\tilde e}^{r}]+$$

$$\frac{1}{4}\,\tilde\mu^2\,\overline{y_e}^{i2p}\,y_e^{i1p}\,\overline{y_u}^{i4r}\,y_u^{i3r}\,\mathrm{LF}_{2,1,1,0}[\tilde\mu,m_{\tilde e}^{p},m_{\tilde u}^{r}]\Big)$$